

CHER SCARLETT

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EDUCATION

2026 **ORANGE COAST COLLEGE** GPA: 4.0

May | A.S., Physics, Mathematics, Astronomy, & Natural Science; A.A. Liberal Arts

2024 **Extracurriculars:** Honors, Astronomy Club (*Treasurer*), STEM Club (*Outreach Manager*),
January MESA, EOPS, Math & Science Club (*President*), $\Phi\Theta\K$, NLSL, SXS group
Professional Memberships: Institute of Physics, Society of Physics Students, American Physical Society, American Geophysical Union, American Mathematical Society, American Association of Physics Teachers, American Meteorological Society

Includes enrollment in other California community colleges.

SKILLS

Python | Java | Groovy | Ruby | PHP | C | JavaScript | CSS | HTML | IDL | Fortran
LaTeX | Mathematica | MATLAB | Jupyter Notebooks | AstroImageJ | ArcGIS | PyTorch

RESEARCH EXPERIENCES

2026 **CUESTA COLLEGE**

Double Star Astrometry | Advisor: Dr. James Eickemeyer

2025 **INSTITUTE FOR STUDENT ASTRONOMICAL RESEARCH (INSTAR)**

Las Cumbres Observatory (LCO) Network

Astronomy Research Seminar | Advisor: Dr. Rachel Freed

2025 **NASA JET PROPULSION LABORATORY (NASA-JPL)**

NASA Juno Mission

Analysis and Archiving of Near- and Mid-Infrared Observations of Outer Planet Atmospheres

Planetary and Exoplanetary Atmospheres Group

Student Independent Research Internship (SIRI) | Advisor: Dr. Glenn S. Orton

2025 **CALIFORNIA INSTITUTE OF TECHNOLOGY (CALTECH)**

| *NASA-JPL Lunar Trailblazer Mission*

2024 *Planetary Sciences Department*

Research Assistant, Science Communication Intern | Advisor: Dr. Bethany Ehlmann

2025 **ORANGE COAST COLLEGE**

Compact Objects and Gravitational Waves | Independent Study | Advisor: Prof. Jessica Asbell

ACADEMIC & WORKSHOP EXPERIENCES

2025 **NASA AI/ML SCIENCE AND TECHNOLOGY INTEREST GROUP**

PyTorch and Auto Differentiation Lecture

2025 **COMPUTATIONAL RESEARCH ACCESS NETWORK | BRYN MAWR | NSF**

Computational Research Methods for Plasma Physics & Sciences Virtual Seminar

Magnetohydrodynamics with FLASH, Monte Carlo Method, & Particle-in-Cell Tracks

2024 **UNIVERSITY OF ILLINOIS | GRAVITY THEORY TRUST**

EFTs, Gravity, and Fluid Dynamics: Progress, Challenges, and Emerging Opportunities Workshop

- 2024 CALTECH | NASA EXOPLANET SCIENCE INSTITUTE (NEXSCI)**
Sagan Exoplanet Summer Workshop
Advances in Direct Imaging: From Young Jupiters to Habitable Earths
- 2024 MIAMI UNIVERSITY | NSF | EARTHSCOPE CONSORTIUM**
Seismology Skill Building Workshop
- 2024 QUEEN'S UNIVERSITY | CERN**
The 6th Summer Particle Astrophysics Workshop (EIEIOO)
- 2024 ARIZONA STATE UNIVERSITY | NASA**
Lucy Student Pipeline Accelerator and Competency Enabler (L'SPACE) Academy
NASA Proposal Writing and Evaluation Experience (NPWEE)
- 2024 NASA**
NASA Community College Aerospace Scholars (NCAS)
- 2003 FRED HUTCHINSON CANCER CENTER | UNIVERSITY OF WASHINGTON**
HutchLab | High School Human Genome Project

OBSERVATION EXPERIENCES

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|-------------|---|-----------------------------|
| 2025 | Caltech Palomar Observatory - Hale Telescope
<i>4 nights; 5.1 m; 1 - 2.5 μm (PHARO)</i>
<i>Observing Jupiter in Support of Juno Mission (NASA-JPL)</i> | <i>Palomar Mountain, CA</i> |
| 2025 | Las Cumbres Observatory (LCO) - Teide Observatory
<i>2 nights; 0.35 m; 0.4 - 0.7 μm (QHY600)</i>
<i>Astrometry of double stars</i> | <i>Tenerife, ES</i> |
| 2025 | LCO - South African Astronomical Observatory
<i>1 nights; 0.35 m; 0.4 - 0.7 μm (QHY600)</i>
<i>Astrometry of double stars</i> | <i>Sutherland, ZA</i> |
| 2025 | NASA Infrared Telescope Facility (IRTF)
<i>10 nights; 3.2 m; 0.8 - 5.3 μm (iSHELL, SPEX, & MIRSI)</i>
<i>Observing Jupiter in Support of Juno Mission; Volatiles in Comet Comae</i> | <i>Mauna Kea, HI</i> |
| 2025 | Lockheed Martin Ignite Ground System (IGS)
<i>4 nights; 8-9 kHz</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>Waterton, CO</i> |
| 2025 | Fukui University of Technology (FUT) Awara Space Center
<i>2 nights; 8-9 kHz</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>Fukui, JP</i> |
| 2025 | Australia Telescope National Facility
<i>6 nights; 8-9 kHz (Mopra)</i>
<i>Monitoring for Lunar Trailblazer (LTB) Signal</i> | <i>New South Wales, AU</i> |

TEACHING & TUTORING EXPERIENCES

2026 August May	EARTHSOPE CONSORTIUM MIAMI UNIVERSITY <i>Intern</i> <i>Seismology Skill Building Workshop</i>	<i>Miami, OH</i>
2025 September May	EARTHSOPE CONSORTIUM MIAMI UNIVERSITY <i>Teaching Assistant</i> <i>Seismology Skill Building Workshop</i>	<i>Miami, OH</i>
2025 May 2024 May	ORANGE COAST COLLEGE PLANETARIUM <i>Presenter</i>	<i>Costa Mesa, CA</i>
2018 May January	UNIVERSITY OF WASHINGTON <i>Adjunct Instructor</i> <i>Introductory Web Development (Extension)</i>	<i>Seattle, WA</i>

ENGINEERING EXPERIENCES

2023 April 2022 March	CONTROLZEE, INC <i>Senior Software Engineer</i>	<i>San Jose, CA</i>
2021 November 2020 April	APPLE, INC <i>Principal Software Engineer, Global Security Tools</i>	<i>Cupertino, CA</i>
2020 April 2019 April	WEBFLOW <i>Senior Software Engineer, Design System & Core Engine</i>	<i>San Francisco, CA</i>
2019 April 2017 November	STARBUCKS COFFEE COMPANY <i>Lead Software Engineer, Web & Mobile</i>	<i>Seattle, WA</i>
2017 August 2016 August	WORLD WIDE TECHNOLOGY <i>Senior Software Engineer</i>	<i>St. Louis, MO</i>
2016 August 	ACTIVISION BLIZZARD <i>Software Engineer, Web & Mobile</i>	<i>Irvine, CA</i>

2015		
August		
2015	USA TODAY	McLean, VA
August	Senior Developer, New and Emerging Technologies	
2011		
November		
2011	NEUTRINO, INC	Seattle, WA
November	Front End Developer	
2008		
January		
2008	LUXURYREALESTATE.COM	Seattle, WA
January	Front End Developer	
2007		
May		

VOLUNTEER EXPERIENCES

2023	LABOR TECH RESEARCH NETWORK	St. Louis, MO
	Board Member	
present		
2021	COWORKER SOLIDARITY FUND	Annapolis, MD
October	Committee Member	
present		
2025	MESA STEM DAY	Newport Beach, CA
April	Workshop Assistant	
2024	MESA CAMPUS COUNCIL	Newport Beach, CA
December	Panlist	
2024	DEMOCRATIC NATIONAL COMMITTEE	Washington, DC
November	Phone Bank Operator	
2024	NEWPORT-MESA COLLEGE & CAREER FAIR	Costa Mesa, CA
October	Booth Operator for Coastline College & MESA	
2024	EXPLORE CALTECH	Pasadena CA
September	Booth Operator for Pasadena City College Physics & Astronomy	
2024	BOLSA CHICA CONSERVANCY	Huntington Beach, CA
May	Invasive Species Removal	
2024	SAN DIEGO COAST KEEPER	San Diego, CA
May	Water Quality Testing	
2023	SURFRIDER FOUNDATION	San Clemente, CA
August	Beach Cleanup	

2015	ORANGE COUNTY PROBATION DEPARTMENT	<i>Santa Ana, CA</i>
<i>August</i>	<i>At-risk youth mentor (OC GRIP)</i>	
2015	MINDMAKERS	<i>Pasadena, CA</i>
<i>August</i>	<i>Youth mentor & web developer</i>	

AWARDS & SCHOLARSHIPS

2026	AMERICAN INSTITUTE OF PHYSICS	<i>Silver Springs, MD</i>
	<i>TEAM-UP Together Research Fellow</i>	
2026	WELLESLEY COLLEGE	<i>Wellesley, MA</i>
	<i>Elizabeth Kaiser Davis Scholar</i>	
2026	NATIONAL OCEANIC & ATMOSPHERIC ADMINISTRATION	<i>Silver Springs, MD</i>
	<i>Ernest F. Hollings Scholar</i>	
2026	JACK KENT COOKE FOUNDATION	<i>Lansdowne, VA</i>
	<i>Transfer Scholar Semifinalist</i>	
2026	COASTLINE COLLEGE	<i>Fountain Valley, CA</i>
	<i>Honor's List (Spring)</i>	
2026	SANTA BARBARA CITY COLLEGE	<i>Santa Barbara, CA</i>
2025	<i>President's Honor Roll</i>	
2025	COASTLINE COLLEGE	<i>Fountain Valley, CA</i>
	<i>President's List (Fall)</i>	
2025	WOMEN'S INDEPENDENCE SCHOLARSHIP PROGRAM	<i>Wilmington, NC</i>
	<i>Doris Buffett Independence Scholar</i>	
2025	AMERICAN INSTITUTE OF PHYSICS	<i>College Park, MD</i>
	<i>TEAM-UP Together Scholar</i>	
2025	COASTLINE COLLEGE	<i>Fountain Valley, CA</i>
	<i>Honor's List (Summer)</i>	
2025	ORANGE COAST COLLEGE	<i>Costa Mesa, CA</i>
	<i>Garrison Fellow</i>	
2025	COMMUNITY COLLEGE LEAGUE OF CALIFORNIA	<i>Sacramento, CA</i>
	<i>All-California Academic Award, First Team</i>	
2025	COASTLINE COLLEGE FOUNDATION	<i>Fountain Valley, CA</i>
	<i>Associated Student Government Career & Technical Merit Scholarship</i>	
2025	COASTLINE COLLEGE FOUNDATION	<i>Fountain Valley, CA</i>
	<i>Herman Tate Endowment</i>	
2025	JEANNETTE RANKIN FOUNDATION	<i>Athens, GA</i>
2024	<i>Jeannette Rankin National Scholar</i>	
2025	NATIONAL SCIENCE FOUNDATION	<i>Alexandria, VA</i>
2024	<i>S-STEM Scholar (Coastline College)</i>	

2024	COASTLINE COLLEGE <i>Honor's List (Fall)</i>	Fountain Valley, CA
2024	ORANGE COAST COLLEGE FOUNDATION <i>Peter Christian Hernandez Memorial Scholarship Award</i>	Costa Mesa, CA
2024	ORANGE COAST COLLEGE <i>Honor's List (Spring)</i>	Costa Mesa, CA
2024	GOLDEN WEST COLLEGE <i>Honor's List (Spring)</i>	Huntington Beach, CA
2022	HACKER NOON <i>Developer Defying Patriarchy (While Being an Awesome Developer) (Runner-up)</i>	Edwards, CO

PUBLICATIONS

- Scarlett, C. (2026). Finding Physicists: The vital role of Community Colleges. *The SPS Observer*, 60(1).
<https://students.aip.org/observer/volume-60-number-1>
- Scarlett, C. (2025). Lunar Trailblazer Interactive Model [Computer software]. *California Institute of Technology*.
<https://trailblazer.caltech.edu/spacecraft/index.html>
- Scarlett, C. (2025). Water into Doom: Balancing AI with Climate Action & Environmental Costs at the Water-Energy Nexus. *The Honors Transfer Council of California Building Bridges Research Conference*.
- Scarlett, C. (2025). The Climate and Tornadoes are Changing. Can Understanding Climate-Driven Chaos Save Lives?. *Academia*.
https://www.academia.edu/150281326/The_Climate_and_Tornadoes_are_Changing_Can_Understanding_Climate_Driven_Chaos_Save_Lives
- Scarlett, C. (2025). Bullshit at the Agora: On the Dangers of a 'Marketplace of Ideas'. *Academia*.
https://www.academia.edu/146050292/Bullshit_at_the_Agora_On_the_Dangers_of_a_Marketplace_of_Ideas
- Scarlett, C. (2025). No Sleep 'Til Payday: Class, Work, and the Price of Survival. *Academia*.
https://www.academia.edu/145993371/No_Sleep_Til_Payday_Class_Work_and_the_Price_of_Survival
- Scarlett, C. (2025). Emmy Noether: A Woman on the Edge of [Space]Time. *Academia*.
https://www.academia.edu/144259985/Emmy_Noether_A_Woman_on_the_Edge_of_Space_Time
- Scarlett, C. (2024). Man Made: Technology, Pygmalion displacement, and the patriarchal idealization of women in science fiction. *Academia*.
https://www.academia.edu/145993413/Man_Made_Technology_Pygmalion_displacement_and_the_patriarchal_idealization_of_women_in_science_fiction
- Scarlett, C. (2024). Remote seismology correlation and Mount St. Helens volcanology. *Seismology Skill Building Workshop*.
https://nbviewer.org/urls/www.iris.edu/hq/files/short_courses/2020/ssb/mt-rainier-seismicity.ipynb
- Scarlett, C. (2024, July 29). *A case of the blues: Is Overexposure to Artificial Light Causing a Mental Health Crisis?*. Esri. <https://storymaps.arcgis.com/stories/e236a32458bb4b7db87171f6a0305d8a>

- Carlson, G., Roginsky, J., & Scarlett, C. (2022, February 6). Washington can be a leader on protecting sexual harassment, assault survivors. *The Olympian*.
<https://www.theolympian.com/opinion/op-ed/article258066068.html>
- Scarlett, C. (2019, March 12). Creating a Spotify-powered app using Nuxt.js. *Smashing Magazine*.
<https://www.smashingmagazine.com/2019/03/spotify-app-vue-nuxt-javascript/>
- Scarlett, C. (2008). sIFRvault [Computer software].